



NATIONAL TEXTILE UNIVERSITY

Department of Computer Science

Lab # 6: Programming Fundamentals (CSC-1071)

Basic Information	
Tools:	Dev-C++ 5.4.1
Objectives	1. Function without arguments 2. Function with arguments
Note	Solve the following problems using the concepts we have covered so far

Function without arguments:

1. Implement three function which displays 20 *, \$, and @. Call these functions inside your program.
2. Write a function which input 5 numbers and display minimum and maximum
3. Write a function which calculate average of five integer after input from user
4. Write a function which uses a nested loop and display triangle of # of size 10 but only display odd rows
5. Write a function which input a number and finds whether it is even OR odd
6. Implement a function named TenSpace which on call displays ten spaces on console
7. Write a function, which prints a triangle of 10 asterisks. Use this function in main five times
8. A company wants to send a four digit number in encrypted form on a network. Write functions with name encrypt and decrypt. The function encrypts input a four digit number and then swaps the first and third; second and fourth digits. The function decrypted input a four digit number and decrypt number in its original form. Force the user to only enter a four digit number.

Function with arguments:

1. Write a function that takes two integers as arguments and returns their sum. Call this function in your main program and display the result.
2. Implement a function that takes an integer as an argument and returns true if it is prime, and false otherwise. In the main program, take an integer input from the user and call this function to check if it is prime.
3. Write a function that accepts two integers as arguments and returns the larger of the two numbers. Call this function from the main program and display the result.
4. Create a function that takes a character and an integer as arguments, and displays the character repeated the specified number of times. For example, calling the function with arguments '*' and '5' should display '*****'.
5. Implement a function that accepts the radius of a circle and calculates and returns the area. Use this function in the main program to take user input and display the area.

6. Create a function that takes two integers and a character (operator) as arguments and performs the specified arithmetic operation (addition, subtraction, multiplication, or division). Call this function in the main program with user inputs.
7. Write a function that takes an integer as an argument and returns its factorial. In the main program, call this function with user input and display the factorial result.
8. Implement a function that takes a year as an argument and returns true if it is a leap year, and false otherwise.
9. Create a function that accepts a character and an integer for row count and displays a square grid of the specified character with the specified row and column size. For example, calling with `#` and `5` would print a 5x5 grid of `#`.
10. Implement a function that takes two dates (day, month, year) as arguments and returns the number of days between them. Use this function in the main program to calculate and display the difference between two user-entered dates.